

Frame transport memo — archive

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Append-only history for `2026-08-15-c912-frame-transport-memo.tex`. Superseded routes, closed checks, and correction trails are kept here verbatim so the live memo can state current conclusions only. Cross-references to statements that remain in the live memo are rendered as bracketed label names.

1 The scissors: hypotheses with no common instance

Archived from the live memo, lines 392–421.

The scissors: these hypotheses have no common instance in the application

Theorem [thm:transport] is true, and in the application it is empty. Its two hypotheses pull in opposite directions.

- If $\Gamma = 0$, so $\mathcal{H} = \Omega_V((z))$, then Levelt–Turrittin supplies solution matrices, but the pro-Laurent comparison gauge has unbounded negative z -order and is not in $\mathrm{GL}_n(\mathcal{H})$.
- If $\Gamma \neq 0$ with the draft’s coefficient-dominant order, the gauge lies in $\mathcal{H}$, but $\mathcal{H}$ is not a formal Laurent series field over its constants and Levelt–Turrittin is unavailable. Concretely, the e_z -component of the Hahn valuation is not a valuation on $\mathcal{H}$: for $f = x^{(0,0)} + x^{(1,-100)}$ and $g = -x^{(0,0)}$ its values on f and g are 0 while its value on $f + g$ is -100 . So there is no Newton polygon, no slope filtration and no Turrittin algorithm. Nor can one descend to the bounded- z -order elements: $f = 1 - zx^{-\gamma}$ has inverse $-\sum_{k \geq 1} z^{-k} x^{k\gamma}$, so they do not form a field.

An abstract Picard–Vessiot ring for a module over $\mathcal{H}$ does exist, with constants $\mathcal{C}$ after passing to the divisible hull. So the missing ingredient is *not* a solution algebra. It is a canonical Levelt–Turrittin normal form over $\mathcal{H}$, singling out which element of the differential Galois group deserves to be called the turn. That is the precise shape of the remaining problem, and it also shows why a purely valuation-theoretic argument cannot supply it.

2 The receiver: tensor product, and a better receiver

Archived from the live memo, lines 433–462.

Why the tensor product cannot simply be dissolved

The map $a \otimes h \mapsto a \cdot h$ into the Hahn field with $\Omega_{V,j}$ coefficients is not well defined: the copy of $H_{0,j}$ inside $\Omega_{V,j}$ consists of degree-zero constants, the copy inside $H_{\text{tot},j}$ of monomial series, and the tensor product exists precisely to identify them. Nor can the two copies be assigned different jobs, with module coefficients read as constants and gauge bookkeeping as monomials: the gauge relation $\Omega' = G\Omega G^{-1} - \theta(G)G^{-1}$ ties G and Ω through the same Novikov variables, so one copy is forced. With the constants reading the gauge leaves the field; with the monomial reading the exponential factors become Γ -supported and Remark [rem:constants-fails] applies.

Two things should also be said about $\mathcal{R}_j$ itself. It is a tensor product of two field extensions and need not be a domain, and the draft reads characteristic polynomials and algebraic multiplicities in $\mathcal{R}_j[X]$; that deserves an explicit remark, which the draft partly gives by proving each factor injects.

A better receiver

Extend the Γ -valuation of $H_{0,j}$ to its algebraic closure and then to $\Omega_{V,j}$, and take the field of series $\sum_k c_k z^k$ with $c_k \in \Omega_{V,j}$ whose support $\{(v_\Omega(c_k), k)\}$ is well ordered in the chosen order. This has a single copy of the coefficients, is a field rather than a tensor product of two fields, hosts the pro-Laurent gauge, and hosts the exponential factors as honest units. It is strictly better than (4.4c). It does not by itself supply Levelt–Turrittin, and it does not remove the rate criterion below.

3 The iterated receiver

Archived from the live memo, lines 544–668.

4 The iterated receiver

The criterion above is a consequence of one architectural choice, and a reader of the second version of this memo proposed the alternative: the draft puts the intrinsic Novikov variables of the endpoint and the new completion variables into a single ordered group, which is what gives an eigenvalue difference positive weight. Nothing forces that. Treat the intrinsic Novikov fraction field as *coefficients*, and complete only in the new exceptional and bulk directions afterwards.

Two facts make the proposal concrete, and both are visible in the draft.

- *The unbounded negative z -order is bulk-driven, not Novikov-driven.* In the recursion (4.1a) the coefficient G_α has no power of z below $-|\alpha|$, where α indexes the *bulk* parameter η , and the filtration depth is acquired by substituting $\eta \in F^1 B$. A completion in the bulk directions alone therefore already hosts the pro-Laurent gauge; the Novikov directions are not needed for it.
- *With the Novikov directions as coefficients, the pathology disappears and the criterion is satisfied rather than evaded.* Eigenvalues of $c_1 \star$ become constants of weight zero, so $\exp(-\lambda/z)$ is not a unit of the receiver but a genuine exponential symbol, the constant-coefficient hypothesis of Definition [def:solution-algebra] holds, and Remark [rem:constants-fails] evaporates. Since $w(\Delta\lambda) = 0$, the inequality $e w(\Delta\lambda) < \varepsilon$ holds trivially, and the product GP of Section [sec:receiver] converges coefficientwise.

The shifts that do involve the exceptional Novikov root do not force those directions back into the filtration. The rank-one unit twist attached to $-(c-1)\lambda_j$, the fixed divisor shift, and the ambient target's $\tau^\circ = O(q^{-1})$ all enter Lemma 4.5 as its a_0 and a_2° terms, which are required only to be *constant in* the positive filtration, not to lie in it. With the Novikov directions as coefficients, $\exp(-a_0/z)$ is an honest exponential symbol rather than the object the draft must argue about separately.

What the sources actually permit

I have checked this against the pinned sources rather than left it as a question, and the answer is positive for the comparison and negative for the gauge.

Iritani's Theorem 5.18 states Ψ as an isomorphism of $\mathbf{C}[z]((q^{-1/s}))[[Q, \tilde{\tau}]]$ -modules, with the change of variables defined over $\mathbf{C}((q^{-1/(r-1)}))[[Q]]$. So the ring is *Laurent* in the exceptional root and *formal* in the ambient Novikov variables and the bulk. Two consequences.

The comparison survives. Inverting the Novikov monomials and completing in the exceptional and bulk directions is a ring homomorphism out of that ring. Ψ , its inverse, and the two inverse identities are module maps over it, so they base-change along it unchanged, and the framed operators' entries lie in the coefficient field, which injects. Nothing in the transport step needs Iritani's proof, only his statement, so the formality of his ring in Q is not by itself an obstacle.

The gauge does not. The source's own reconstruction writes $\tau(\tilde{\tau}) = \tau^\circ + t(\tilde{\tau})$ and $\varsigma_j(\tilde{\tau}) = \varsigma_j^\circ + s_j(\tilde{\tau})$, with $t(\tilde{\tau}) \in H^*(X)((q^{-1}))[[Q, \tilde{\tau}]]$ and s_j of the same shape over $H^*(Z)$. The initial terms are harmless: τ° and ς_j° are by definition the values at $Q = \tilde{\tau} = 0$, hence carry no Novikov dependence at all, and the exceptional direction remains a completion direction as the draft already has it. But t and s_j vanish only at $Q = \tilde{\tau} = 0$, so at $\tilde{\tau} = 0$ they retain pure-Novikov terms. Those have to be removed by the normalized gauge of Lemma 4.5, and that construction converges by topological nilpotence of the bulk shift. Inverting the Novikov directions destroys exactly that.

So the iterated receiver is not free, and the decisive question is now small and graded rather than general:

Is the pure-Novikov part of the coordinate shift — t and s_j restricted to $\tilde{\tau} = 0$ — confined to $H^0 \oplus H^2$?

If it is, the string and divisor equations absorb it into the a_0 and a_2° terms of Lemma 4.5, which are required only to be constant in the positive filtration, and the iterated receiver goes through for the ambient summand with $w(\Delta\lambda) = 0$. If it is not, the Novikov directions must stay inside the filtration, eigenvalue differences reacquire positive weight, and the criterion of Section [sec:receiver] returns.

The grading answers this, and the answer is negative in general. Iritani's Theorem 5.18(5) states that $\tau(\tilde{\tau})$ and $\varsigma_j(\tilde{\tau})$ are homogeneous of degree 2, and his conventions are $\deg Q^d = 2c_1(X) \cdot d$ and $\deg \tau_i = 2 - \deg \phi_i$. A pure-Novikov term $Q^d \phi$ therefore satisfies

$$\deg \phi = 2 - 2c_1(X) \cdot d,$$

so $c_1 \cdot d = 1$ contributes in H^0 , $c_1 \cdot d = 0$ in H^2 , and $c_1 \cdot d = -1, -2, \dots$ in $H^4, H^6, \dots$; classes with $c_1 \cdot d \geq 2$ cannot contribute at all. Hence the required confinement to $H^0 \oplus H^2$ follows exactly when every contributing effective class satisfies $c_1 \cdot d \geq 0$, and it is not automatic: Iritani's

theorem is stated for an arbitrary smooth projective X , and after several blowups in a weak factorization there is no reason for the intermediate ambient variety to have nef anticanonical class.

So the iterated receiver does not by itself remove arbitrary intermediate fourfolds from the problem. It removes them under a semipositivity hypothesis on the ambient, which the weak factorization does not supply. This is worth stating plainly, because it is the second proposed repair that fails on inspection, and it fails for a reason the draft would have inherited silently.

For the center the same analysis applies, with the additional and unchanged difficulty that its Novikov map is the noninjective one, which is why divisor tagging exists.

The scope consequence is the reason to try this first. If the ambient endpoint is handled this way, no inequality is needed for arbitrary intermediate fourfolds — the case that looked least tractable. What remains is the center summands, and there the draft’s own low-dimensional classification is restrictive: for nef canonical class it produces an integral- z gauge to a regular-singular connection, so there is no exponential separation at all and $w(\Delta\lambda) = 0$ again; the remaining minimal cases are $\mathbf{P}^1$, $\mathbf{P}^2$, ruled surfaces, and point blowups. Proving primitive-sixth vanishing directly for those, in the coordinates Iritani’s specialization actually produces, is a smaller problem than a general transport theorem.

5 The framing route as proposed, and checks one to three

Archived from the live memo, lines 669–835.

6 The framing route, checked against the sources and closed

The second version of this memo proposed the following route and called it the probable repair. I state it as it was proposed, then report the three checks, then state what survives. The checks were run against arXiv:2411.02266 as cached, arXiv:2307.03696v4, and arXiv:2307.13555v3; the full verdicts, locators and file hashes are in the companion report *The three framing-compatibility checks, run against Iritani’s decomposition*.

The route as proposed

Everything above treats the pro-Laurent bulk gauge of Lemma 4.5 as unavoidable. It may not be. Hinault, Yu, Zhang and Zhang work with F -bundles on the formal u -disc, where a framing is a regular flat connection and, in a framing-flat trivialization, the meromorphic connection takes the normal form

$$\nabla_{u\partial_u} = u\partial_u + u^{-1}K + (\mu D + H).$$

Their Theorem 4.34 characterizes when two connections of that shape are gauge-equivalent under

$$\Phi(u) \in \mathrm{GL}(H[[u]]), \quad \Phi_{ij}(u) \in k[c_1, \dots, c_r][[u]],$$

with Φ then unique given its value at $u = 0$; the conditions are that the leading operators K be conjugate by a parameter-polynomial ϕ , that the two μ agree, and that the diagonal entries of H agree after conjugation modulo the parameter ideal. Their applications are uniqueness of the decomposition for projective bundles and for blowups, complementing the existence theorems of Iritani–Koto and Iritani.

The shape of that gauge is exactly what this memo has been missing. A gauge in $\mathrm{GL}_n(K[[u]])$ has only nonnegative integral powers of the loop coordinate, adjoins no root of u , and is therefore fixed by the turn $u \mapsto e^{2\pi i}u$. Taking $\Gamma = 0$, so that $\mathcal{H} = \Omega_V((u))$ and Levelt–Turrittin is available, Theorem [thm:transport] applies with no further work:

$$\Phi \in \mathrm{GL}_n(K[[u]]), \quad \nabla' = \Phi \nabla \Phi^{-1} - \theta(\Phi) \Phi^{-1} \implies M_{f,V}(\nabla') \text{ and } M_{f,V}(\nabla) \text{ conjugate.}$$

No Hahn receiver, no product GP , and no inequality $ew(\Delta\lambda) < \varepsilon$, provided the decomposition of Proposition 4.7 can be identified with an F -bundle isomorphism of the class Theorem 4.34 governs. That identification, the matching of the $u = 0$ datum, and the separation of the center specialization were the three checks.

Check one: the identification does not exist as posed

It fails three times over, and the first two are independent of the draft.

Domain. Section 4.4 of Hinault–Yu–Zhang–Zhang is titled *Equivalence of F -bundles over a point*, and both of its applications evaluate at the closed point $q = t = 0$ of the base, the large-radius limit; the proof of their Theorem 5.16 opens by observing that there the quantum product reduces to the classical cup product, so that the leading operator K_{split} of their (5.18) is a cup-product operator, hence scalar-plus-nilpotent on each block rather than nilpotent outright. The draft’s ν_6 is not evaluated there. It is read where the eigenvalues of $c_1\star$ separate — at the cubic, where the block carries $r = (3q)^{1/2}$ and the indicial roots $\pm 1/6$, $\pm 5/6$. At $q = 0$ that separation collapses and the exponential blocks the framed operator is built from do not exist. Their Theorems 5.20 and 5.24 do reconstruct the isomorphism over the whole base from its restriction at the limit point, but that is uniqueness of a map whose existence is Iritani’s and Iritani–Koto’s theorem; it is not a transport of framed monodromy between base points.

Hypotheses, for the blowup. Section 4.4 assumes the generalized eigenspaces of K all have the same dimension, and the matrix calculus of Definition 4.33 and Theorem 4.34 rests on the resulting splitting $H_0^{\oplus m} \cong H$. The blowup decomposition is $H^*(X) \oplus \bigoplus_{i=1}^{m-1} H^*(Z)[-2i]$, whose summands differ in dimension whenever $\dim H^*(Z) \neq \dim H^*(X)$; the authors write a general endomorphism there as a matrix with $\Phi_{1,1} \in \mathrm{End}(H^*(X))$ and $\Phi_{i,i} \in \mathrm{End}(H^*(Z))$, so the mismatch is explicit in their own notation. Their blowup statement, Theorem 5.22, does assert existence of the isomorphism, and Theorem 5.24 gives the corresponding uniqueness; the point is not that it is unproved but that it is not *derived from* Theorem 4.34, whose hypotheses do not cover the unequal block sizes, and that its existence is referred to the earlier decomposition work. So Theorem 5.22 supplies no leverage of its own on the blowup half of Proposition 4.7 — equation (4.3), the half the headline theorem cannot do without — beyond what Iritani’s theorem already gives. The projective-bundle half is genuinely covered: there the eigenspaces are m copies of $H^*(X)$ by their Lemma 5.8, matching Section 4.4, and Theorem 5.16 is proved.

Purpose. Even where Theorem 4.34 applies, its conclusion compares the source at its base point with the target based at a shifted class $\Delta(a)$, pinned by their (5.17) for projective bundles and (5.23) for blowups. That shift is exactly what the pro-Laurent gauge of Lemma 4.5 exists to undo. The route relocates the bulk displacement rather than removing it.

The two normalizations do agree on where the displacement sits, which is a useful check on the draft’s reading of the sources. In their limit-point coordinates the eigenvalue $\lambda_i \in \mathbf{C}[c_1, \dots, c_m]$ is cohomology-valued, so $\Delta(a)$ acquires components in $H^{\geq 4}$ from the higher Chern classes, while the H^2 components cancel identically — their Lemma 5.8(2) gives $m\lambda_i \equiv m\xi^{i-1} - c_1V$, cancelling the c_1V on the right of (5.17) — and the H^2 part of $\Delta(a)$ is left free. In Iritani’s coordinates the

same data reads $\varsigma_j^\circ = -(c-1)\lambda_j + h_{Z,j} + O(q^{-1/(c-1)})$ with λ_j a *scalar* multiple of $q^{1/(c-1)}$, so the H^0 part is that scalar, the H^2 part is $h_{Z,j}$, and everything of degree at least four has moved into the tail. Same content, different bookkeeping; the draft's split of ς_j° into unit twist, fixed divisor, and positive-filtration tail is right. The tail is present in general, but *not* always: an earlier version of this passage inferred from the $O(q^{-1/r})$ symbol that even $X \times \mathbf{P}^1$ carries a tail and so does not escape the bulk gauge. That inference was unsupported — the $O(\cdot)$ symbol carries no information about $[z^{-1}]\log$ either way — and it is contradicted by Theorem [thm:trivial-displacement] below, which shows that for a trivial bundle $\varsigma_j^\circ = r\lambda_j$ *exactly*, with no H^2 term and no $H^{\geq 4}$ tail. The claim is struck.

Check two: the $u = 0$ datum, which passes

Their Theorems 5.20(2) and 5.24(2) determine the base map from its restriction to the base point up to a multiplicative constant in the q direction (the statement for all logarithmic directions is their Proposition 4.31(2)); the proof of their Proposition 4.31 shows why, since df is recovered through $\Psi(d \log p_i) = d \log f_i$ and integrating a logarithmic form leaves one constant. That ambiguity is a character of the Novikov lattice: $q \mapsto cq$ is $Q^d \mapsto c^{E \cdot d} Q^d$, which is the divisor substitution (4.1) of the draft with a_2° a logarithmic multiple of the dual class. The draft already proves such a substitution is a z -constant coefficient automorphism and leaves ν_6 unchanged, and already checks it is well defined on the image monoid. The bundle map carries no ambiguity at all once its restriction is fixed.

The ambiguity also never has to be tolerated, because the sources pin it. Iritani–Koto's (5.11) gives $\varsigma_j^\circ = r\lambda_j + [z^{-1}] \log(q^{c_1(V)/(rz)} F_j(1))$ in closed form and their (5.12) gives Φ_j° ; Iritani's Section 5.8.1 fixes $\Psi|_{Q=\tilde{\tau}=0}$ from his Lemma 5.13 and the explicit λ_j . One manuscript sentence citing those, and noting that any residual constant is a divisor substitution already covered by (4.1), settles this point for good, whichever repair eventually lands.

Check three: the center specialization, which stays separate

It stays separate more firmly than the check anticipated. Their A -model F -bundle is built throughout on the one-variable Novikov projection $q^\beta \mapsto q^{\beta \cdot \omega}$ of their (2.21), and in the blowup application the center copies are the maximal F -bundles of Z associated to $\sigma^* \omega_X$ — the Novikov direction pulled back from the ambient variety, which is the draft's specialization in a coarser form. No statement of theirs can therefore certify that a specialized center module carries an intrinsic framed monodromy. Divisor tagging remains the route, in the current manuscript, for identifying the specialized center contribution with the intrinsic invariant; route 3 of the list below bypasses that identification instead, by proving specialized vanishing directly in low dimension.

One genuine gain came out of it. Their Lemma 2.24 proves that the collapsed potential determines the full genus-zero potential, by expanding in the H^2 bulk directions and separating curve classes by their intersection numbers with an H^2 basis, under the finiteness of their Assumption 2.22. That is the mechanism of `lem:divisor-tagging`, in print, in a closely related setting. It does not close that lemma — it works at the level of the potential over a base retaining all H^2 directions, whereas the draft needs the separation at the level of a framed operator at a fixed bulk point, and the draft's collapse is by the pair $(i_*, \rho_C \cdot)$ rather than by a single nef class — but it makes the shape of the argument a cited one.

7 The corrected program (superseded by the rigidity section)

Archived from the live memo, lines 1121–1150.

The corrected program

1. Flatness preserves the spectral decomposition of U and gives the exact evolution $D_a U_i = C_{a,i} + [C_{a,i}, \mu_i]$, unconditionally.
2. Multiplicity-one blocks contribute no cyclotomic monodromy (Theorem [thm:simple-blocks]), so only coalesced blocks are in play.
3. For a coalesced block the off-block projector equations are governed by the full Sylvester operator $(u_j - u_i) + L_{N_j} - R_{N_i}$, invertible but not scalar.
4. What must be deformed is the entire formal block normal form $J + zD + z^2E + \cdots$, not the compressed pair, and the deformation theory has to reproduce the cubic 2×2 example above.

A caution on the literature before it is leaned on. The extension results for isomonodromic deformations with coalescing eigenvalues assume, among other hypotheses, that the leading matrix is diagonalizable at the coalescence point. The carrier here is nonsemisimple at leading order — $J_0 \neq 0$ — so those theorems do not apply off the shelf, and the coalescence question stated below is not answered by citing them.

The risk that remains is unchanged in kind and now better located: a coalesced block can *split* under the displacement, the splitting being governed by $D_a U_i$, and if the rank-two block carrying $\pm 1/6$ split into simple blocks those would carry residue 0 by Theorem [thm:simple-blocks] and ν_6 would drop by two. Proving that the reconstruction displacement does not do that, or does it in a count-preserving way, is the residual problem. Section [sec:rigid] now does exactly that, by shearing the block before deforming it.

8 A shorter route to the endpoint via the Serre decoration (not taken)

Archived from the live memo, lines 1542–1740.

A shorter route to the endpoint, in the sources’ own terms

Two paragraphs of Katzarkov–Kontsevich–Pantev–Yu’s Example 6.21 bear directly on this and should be on the record, because between them they suggest a route to the one-stabilization theorem that needs neither this memo’s transport lemma nor its rigidity theorem.

First, they compute the atomic composition of a smooth cubic threefold at the hyperplane point: two one-dimensional atoms for the nonzero eigenvalues of $\text{Eu} \star$, and one atom $\alpha(X)$ for the eigenvalue 0. That is the block structure used throughout this memo, in their language. They add that “the Witt algebra argument from Remark 3.14 shows that there is no further splitting in a neighborhood of b ” — which is Theorem [thm:no-splitting], asserted rather than proved, and their Remark 3.14 introduces its finer clauses with “more fine results can be proved”.

Section [sec:rigid] supplies proofs of both the splitting statement and the finer one, for this block type, from flatness alone.

Second, and more consequentially, they separate $\alpha(X)$ from a curve atom not by formal monodromy but by a Serre automorphism. They observe that $\alpha(X)$ looks like the atom of a smooth projective genus-five curve, then note that as a Hodge atom enhanced with its Serre automorphism S , the graded minimal polynomial forbids S from having eigenvalue -1 , so it cannot be the Serre automorphism of a genus-five curve. They conclude that every smooth cubic threefold is irrational, and remark that the same approach applies to other Fano threefolds.

Their $S^5 = [3]$ is quoted correctly above — it is what Example 6.21 says — but it is not the relation to use when computing on a K-group. Kuznetsov’s fractional Calabi–Yau relation for the same category is $S^3 = [5]$, and on the numerical Grothendieck group, where a shift $[k]$ acts by $(-1)^k$, the two predict $S^3 = -I$ and $S^5 = -I$ respectively; the Serre operator computed from Riemann–Roch satisfies the first exactly and fails the second, its characteristic polynomial being Φ_6 and its order six, in agreement with the exponents $-1/6, -5/6$ recorded after [eq:sheared-data]. The offset is systematic rather than a slip: their Example 6.19 gives $S^3 = [4]$ for a very general quartic in $\mathbf{P}^5$, whose Kuznetsov component is a K3 category with $S = [2]$ categorically. (An earlier version of this passage cited an “Example 6.20” about a cubic fourfold; no such example exists, and the statement belongs to Example 6.19 and to the quartic.) The two are different objects — the Serre automorphism of a Hodge atom in the cohomological $\mathbf{Z}$ -grading, and the categorical Serre functor on a K-group — so a citation should follow the object it is about. Locators and the full comparison are in ‘2026-08-15-c912-kkpy-imports-source-check.md’.

A second normalization difference runs alongside it and explains a substitution this lane had observed but not accounted for. Their statements are made after the half-parity gauge u^g of Claim 6.15, this memo’s count before it. That gauge shifts exponents by half the cohomological degree, multiplying monodromy eigenvalues by a parity-dependent sign: primitive sixth roots become primitive cube roots and ± 1 swap. It is the $\lambda \mapsto -\lambda$ substitution between the prime-Fano classification’s polynomial R and the Serre side. Every separation used below distinguishes roots of unity of order at most two from roots of order three or six, which the gauge preserves.

The consequence for the endpoint is worth stating plainly. Their argument is dimension three, where Proposition 5.30 requires excluding only points and curves. The one-stabilization statement is dimension four, where the same proposition requires excluding points, curves *and surfaces*. Everything else — the projective-bundle formula putting two copies of $\alpha(X)$ in $X \times \mathbf{P}^1$, the non-rationality criterion, and the decoration itself — is already theirs. So the endpoint theorem may reduce to one classification question, which in the terms of this memo reads:

(iii) Every atom of a smooth projective surface has formal monodromy of order at most two, its eigenvalues lying in $\{+1, -1\}$. In particular no such atom has a primitive sixth root of unity as an eigenvalue, and no Serre operator on such an atom’s numerical Grothendieck group has Φ_6 as a factor.

The stronger form is the one Katzarkov–Kontsevich–Pantev–Yu’s own dimension-four example uses, where the exclusion is unipotency against non-unipotency. Their Example 6.19 needs it only for surfaces of general type, the atom there being shaped like one; the endpoint needs every surface, because the cubic threefold’s atom is shaped like a genus-five curve atom and nothing narrows the search in advance.

Two reductions come with the statement, and one warning.

Rank-one atoms are automatic. For a rank-one lattice the Euler form is a 1×1 matrix, hence symmetric, so $S = E^{-1}E^T = 1$. A semisimple small quantum cohomology has every block

of rank one, so (iii) holds for every surface with semisimple small quantum cohomology with nothing to prove. For a surface with nef canonical class their Lemma 5.24 gives a single atom and Claim 6.15 gives a regular singularity with nilpotent residue after the half-parity gauge, putting every eigenvalue at a root of unity of order at most two.

The remaining surfaces, by the minimal model program. These two reductions leave a short induction, which closes (iii) modulo three imported statements.

Remark 1 (This is Theorem [thm:low-dim], and the two readings coincide). Statement (iii) is not new work: it is Theorem [thm:low-dim], proved in Section [sec:endpoint] by exactly this induction — blow down to a minimal model, use their Lemma 5.24 and Claim 6.15 for nef canonical class, and reduce $\mathbf{P}^2$ and the geometrically ruled surfaces to point and curve atoms through the projective-bundle identity. What the present section adds is only that the Serre-decorated reading of (iii) is the same statement rather than a companion to it: Katzarkov–Kontsevich–Pantev–Yu *define* the Serre automorphism as the monodromy in the u -direction, with $\chi(a, b) = \chi(b, S(a))$, so an atom’s Serre eigenvalues are its formal monodromy eigenvalues and the shorter route’s step (iii) is discharged by Theorem [thm:low-dim] as it stands.

Two refinements are worth recording. First, the conclusion is sharper than $\nu_6 = 0$: in every case the proof produces monodromy of order at most two, which is the discriminator their own dimension-four example uses. Second, the blowup input can now be taken from Iritani’s Theorem 5.18 directly, an isomorphism of quantum D -modules commuting with the connection and preserving the Euler vector fields, rather than only through the chemical-formula identity.

The fake of the previous paragraph does not disturb the blow-down step: it mixes the new exceptional object with the ambient structure sheaf, which lie in different atoms, so the pair it generates is not an atom. Nor is the blow-down step resting on a multiplicity count: the computational literature — Böhning, von Bothmer and Su’a — verifies the blowup statement for multiplicity sequences and records that naive and small atomic decompositions can differ, but Iritani’s theorem is at the level of the connection itself, which is the reading used above. The reading list, cache locators and hashes are in ‘2026-08-15-c912-blowup-formula-source-check.md’. Iritani also records that Gyenge and Szabó studied the asymptotics of the Euler eigenvalues under blowdowns to minimal models of surfaces, which is the configuration this induction runs through.

The ambient category never sees it. The Serre functor of a surface is tensor by ω followed by $[2]$, and $[2]$ acts by $+1$ on K -groups, so the ambient numerical Serre operator is multiplication by $\exp(K)$, which is unipotent. A primitive sixth eigenvalue on a surface can only come from a decomposition, never from the surface’s own K -theory.

The warning: admissible is not enough. For a subcategory generated by an exceptional pair with $\chi(e_1, e_2) = x$ the numerical Serre operator has characteristic polynomial $\lambda^2 + (x^2 - 2)\lambda + 1$, which is Φ_6 exactly when $x^2 = 1$. On any surface carrying a (-1) -curve E — that is, any non-minimal surface — the sheaf $O_E(-1)$ is exceptional, $\mathrm{Hom}^\bullet(O_X, O_E(-1)) = 0$, and $\chi(O_E(-1), O_X) = -1$, using only $E^2 = -1$ and $K_X \cdot E = -1$. So $\langle O_E(-1), O_X \rangle$ generates an admissible subcategory whose Serre operator has order six. Statement (iii) is therefore false if “atom” is weakened to “admissible subcategory”, and no lattice argument ranging over admissible subcategories can prove it. The subcategory is not an atom: it is generated by an exceptional pair and splits into two rank-one pieces whose Serre operators are both the identity. Since the blowup is exactly what supplies this fake, the blowup step of any proof of (iii) is where the care is needed.

With all four imports now read at the source, the route to “ $X \times \mathbf{P}^1$ is not rational” runs in four

steps. Their Example 6.21 gives the atom $\alpha(X)$ at the eigenvalue zero. Its formal monodromy has the two primitive sixth roots of unity as eigenvalues, by the exponents $-1/6, -5/6$ computed after [eq:sheared-data]; that is order six. Theorem 4.11 with $r = 2$, applied to $X \times \mathbf{P}^1 = \mathbf{P}(\mathcal{O} \oplus \mathcal{O})$, puts two copies of $\alpha(X)$ in the atomic decomposition of $X \times \mathbf{P}^1$. And no atom of a smooth projective variety of dimension at most two has monodromy of order six: points give the identity, curves give -1 twice, and surfaces have order at most two by Theorem [thm:low-dim].

The criterion applied at the end is *not* their numbered Proposition 5.30, which excludes membership in HAtoms of low dimension for undecorated atoms, whereas the separation just made uses the monodromy, a decoration. It is the enhanced form, which the same section states as unnumbered prose: if a smooth projective variety has an enhanced atomic composition containing an enhanced atom that does not come from a smooth projective variety of dimension at most $\dim X - 2$, then it is not rational. With $\dim(X \times \mathbf{P}^1) = 4$ the excluded dimensions are at most two, which is what the four steps supply. The numbered proposition is proved there as a special case of their Proposition 5.17; the enhanced version is asserted, on the ground that enhancement by Euler pairings and Serre automorphisms repeats the undecorated theory. So the load-bearing citation of this route is an assertion, and a manuscript using it must either prove the enhanced criterion in the generality it needs or say plainly that it quotes one. The route uses no transport lemma, none of Section [sec:unconditional], and no integral structures.

This is not the same invariant as ν_6 , and it is not obviously weaker or stronger; it is the sources' own, which is an argument for using it. Two cautions. It buys the Hodge-atom, motive and Serre-enhancement machinery that the earlier blueprint deliberately avoided, so the manuscript would import more than the two decomposition theorems. The deferral is wider than an earlier version of this passage recorded: in the non-archimedean analytic setting, which is the one an A -model F -bundle of a projective variety actually lives in, the authors defer the Euler pairing and the Serre automorphism themselves to forthcoming work, not merely the integral-structure enhancement. So the decoration the route turns on is not available off the shelf in the setting required. And their sentence is a worked example, not a theorem with hypotheses, although the statements it invokes — Lemma 5.24, Claim 6.15, Proposition 5.30 and Theorem 4.11 — have now been read at the source and say what is claimed of them here. The graded minimal polynomial has been checked on the lattice as well, and the two conventions differ as described above; the surface exclusion is Theorem [thm:low-dim], with Remark [rem:step-iii-is-low-dim] for why the decorated reading needs nothing further. Backing computations, and the exact record of what was and was not read at the source, are in ‘2026-08-15-c912-step-iii-serre-restatement.md’ and ‘2026-08-15-c912-step-iii-surface-induction.md’, with the scripts ‘2026-08-15-c912-surface-atom-serre-check.py’ and ‘2026-08-15-c912-step-iii-surface-induction-check.py’.

9 Which other routes also close (merged into the routes section)

Archived from the live memo, lines 2177–2259.

Which other routes also close

Three, at different costs.

The draft's own presentation. Keep the operation formulas and transport ν_6 summand by summand. This needs the specialization-safe statement at every intermediate fourfold, so it needs Section [sec:rigid] generalized from a rank-two Jordan block to an arbitrary block. That

generalization is plausible for a single Jordan block of any size by the same shearing, and needs a resonance hypothesis in the semisimple case; it is real work, and it is the only route that keeps the draft’s current architecture.

The sources’ Serre decoration: restricted, referee-risky, and not taken. On its face it avoids the gap of Section [sec:endpoint-gap] by construction, which is what made it look attractive; the reasons not to take it are recorded after the description. Their Proposition 5.23 proves that the isomorphism class of the fibre representation of an atom is independent of the representative, using rigidity of finite-dimensional representations of a proreductive group — a global argument, valid across a connected component, which monodromy data does not enjoy. They then decorate with a Serre automorphism and state that the enhanced theory is a straightforward repeat of the undecorated one, with the corresponding strengthened non-rationality criterion.

Their Example 6.21 separates $\alpha(X)$ from a genus-five curve atom by the graded minimal polynomial $S^5 = [3]$ and concludes that every smooth cubic threefold is irrational. More to the point for the endpoint, their Example 6.17 runs the same pattern in *dimension four*, for a very general cubic fourfold containing a plane: there $\alpha(X)$ is shaped like the atom of a surface of general type, and they exclude it by playing $S^3 = [4]$ against the unipotency that Claim 6.15 forces on the even cohomology of a nef-canonical surface. That is exactly the shape the one-stabilization argument needs, with $X \times \mathbf{P}^1$ in place of their fourfold, and it pairs their globally well-defined decoration on the cubic side with a pointwise structural statement on the surface side — the two halves whose mismatch is the gap above.

Why it is not taken, in order of severity.

The decoration is not defined in the category our atoms live in. They define the duality automorphism S for a *complex analytic* F-bundle of exponential type with nondegenerate pairing, as the monodromy in the u direction, and then say that “the definition in the case of non-archimedean analytic F-bundle requires extra work, and will be discussed in [49]”. But the A-model F-bundle of a projective variety is built over $k((y))$ and base changed to its algebraic closure, and is a non-archimedean analytic F-bundle in their own words. So in the setting where Hodge atoms of a projective variety live, the decoration this route turns on is not merely unproved but undefined as the paper stands. That is not a gap a lemma of ours can close; supplying the definition is the content of their deferred work.

The enhanced theory is asserted, not written out. The sentence that the enhanced theory “is completely straightforward and is just a repeat of the theory of undecorated Hodge atoms” is unnumbered running text. The corresponding strengthened non-rationality criterion, which is the one this route would apply, is stated in the same register; the numbered and proved Proposition 5.30 is the undecorated version, which does not separate atoms by a decoration. A manuscript may not stand on the unnumbered version.

The statements live in three different categories. Their Leray–Hirsch Theorem 4.11 is an isomorphism over analytic domains, their atomic statements hold near a chosen base point, and Iritani’s blowup decomposition is formal over an extended Novikov ring with fractional powers of q . Any argument of ours would stitch formal, non-archimedean analytic and complex analytic statements together, and every stitch is a place a referee asks precisely what was transported.

And the examples are examples. Their Examples 6.17–6.21 are worked cases, not theorems with hypotheses, so the surface exclusion for $X \times \mathbf{P}^1$ would have to be done rather than cited in any case.

One way the first objection could be dodged is to work complex analytically, where S is defined, which requires convergence of the relevant quantum cohomology and so trades Hypothesis

4.7H for a convergence hypothesis. That is a hypothesis of a more familiar kind, but it is still a hypothesis, and the formal treatment of this memo was chosen precisely to avoid needing one. The route is therefore recorded as a restricted and referee-risky alternative with heavy import costs, not as the plan.

Both decorations at once. They are independent invariants of the same atom, so proving the surface exclusion twice is a genuine referee-facing check rather than duplication. If the Serre computation is cheap it is worth having as a second column in the same table.

10 The trade between the two sides, and where it cancels

Archived from the live memo, lines 2372–2428.

The trade between the two sides, and where it cancels

It is worth being explicit about what is being traded, because the accounting localizes the remaining debt more sharply than the narrative does.

Vertical motion is free; horizontal motion is the whole debt. Each step of a factorization gives an exact identity $\nu_6(\mathrm{Bl}_Z V, w) = \nu_6(V, \tau(w)) + \sum_j \nu_6(Z, \varsigma_j(w))$, and with $\dim Z \leq 2$ the centre terms vanish. So moving between varieties conserves the invariant exactly: nothing leaks into a centre and nothing is created. The only way the invariant can change anywhere in the ledger is by moving between two parameters of *one* variety. Every obligation this memo has chased is a horizontal one, and the vertical steps have never been in doubt.

Consequence, in contrapositive form. Suppose $X \times \mathbf{P}^1$ is rational. The left end carries 4 and the right end carries 0, and conservation across every vertical step then forces some valley of the zigzag to have a bulk-parameter-dependent primitive-sixth multiplicity. So the theorem is equivalent to: no smooth projective fourfold arising as a valley has a ν_6 -carrying block that degenerates along the displacement the chain produces. Stated this way the debt is a single negative statement about degeneration, not a positive statement about transport.

The balance has slack, and the slack is three. The left end supplies 4, being two copies of the cubic’s rank-two block, and the contradiction needs only 1. So up to three of the four eigenvalues may be lost anywhere along the chain without harming the argument. What must be excluded is total collapse, not any loss — a weaker statement than constancy, and one that Section [sec:rigid] overshoots, since it proves the characteristic polynomial exactly constant.

Four places where the debt cancels outright.

1. Any step whose accumulated displacement lies in $H^0 \oplus H^2$. The string and divisor equations are exact there, so such a step costs nothing. This is not a rare case: it is exactly what makes the endpoint free, where the displacement is $\pm 2q^{1/2}$ and purely H^0 .
2. An adjacent blowup and blowdown of the same centre, where the composite parameter map is the identity and no comparison of two parameters arises.
3. Any valley whose Euler operator is semisimple, or more generally has only multiplicity-one eigenvalues in the relevant range. There Theorem [thm:simple-blocks] gives $\nu_6 = 0$ at every parameter unconditionally, so the two incoming parameters agree trivially and the valley is vacuous.
4. Every centre, by the low-dimensional lemma. Centres are where a leak would have to go, and they cannot receive one.

What is left after the cancellations. The debt sits only at valleys that simultaneously carry a nonzero ν_6 and receive a displacement with a nonzero $H^{\geq 4}$ part. By the dimension axiom each such displacement contributes finitely much per Novikov degree, so the surviving obligation is finite data at finitely many places, and by the slack it need only be shown not to annihilate all four eigenvalues at once. That is the sharpest available statement of what remains, and it is what a next attempt should target rather than constancy in general.

11 Corrections to the earlier versions of this memo

Archived from the live memo, lines 2570–2712.

12 Corrections to the earlier versions of this memo

From the fifth version to this one

Nothing in the fifth version is retracted; one of its hypotheses is discharged and one of its expectations is corrected.

1. Standing hypothesis (H2) of Section [sec:rigid] is deleted. It is a theorem: Lemma [lem:duality-gauge] shows the decoupling gauge is an isometry of the Poincaré pairing, and Theorem [thm:h2-automatic] then forces $f \equiv 0$ pointwise on the germ, for every rank-two coalesced block of every smooth projective target. Theorem [thm:no-irregularity] and Corollary [cor:cubic-closed] now hold under (H1) alone. The flatness route to $f \equiv 0$ is kept as a remark, since it is what survives without the Frobenius property.
2. The shorter-endpoint-route subsection recorded the graded minimal polynomial as $S^5 = [3]$. On the numerical Grothendieck group that is incompatible with the count: it forces order ten, where the Serre operator computed from Riemann–Roch has characteristic polynomial Φ_6 and order six, matching Kuznetsov’s $S^3 = [5]$. The subsection now uses the verified relation, and states the surface obligation as (iii) with its two reductions and the non-minimal-surface counterexample to the weakened form. It also now records that the shorter route’s step (iii) is discharged by the existing Theorem [thm:low-dim], since their Serre automorphism is defined as the monodromy and the two readings therefore coincide (Remark [rem:step-iii-is-low-dim]); an intermediate version of this memo carried a separate proposition duplicating that theorem, now removed.
3. The scope remark expected the rank-two mechanism to extend to a Jordan block of any size. It half does. Proposition [prop:size-m-steps] carries the commutant and no-splitting steps to size m , but Proposition [prop:size-m-duality] shows duality removes only the deepest of the pole coefficients the general shearing produces, so rank two is special rather than typical, and for $m \geq 3$ the base point must be settled by the grading, whose missing step is named there.

From the fourth version to the fifth

Nothing in the fourth version is retracted. Section [sec:rigid] is added, and three of its consequences change earlier readings.

1. Section [sec:not-yet] said the coalesced case was not reduced and named the block-splitting risk as the residual problem. Both are now settled for a rank-two Jordan block: Theorem [thm:no-splitting] excludes the splitting and Theorem [thm:rigidity] freezes the exponents. The section is left standing because its diagnosis — that the compressed pair is the wrong object — is what the shearing acts on.
2. The residual gap of Section [sec:hyzz] is closed for the cubic ambient summand by Corollary [cor:cubic-closed]. It remains open as stated for arbitrary smooth projective T , which is a statement about arbitrary blocks, not about transport.
3. Cai's Proposition 6 already asserts the $\pm 1/6$ exponents for the big quantum connection, by a bulk gauge of exactly the pro-Laurent shape this memo scrutinized. That argument is sound with the bulk kept formal; the receiver discussion was always about specialization, not about the formal statement. Section [sec:rigid] makes the point moot by producing constants.

From the third version to the fourth

Both are in Section [sec:blockwise], and both are at the coalesced blocks, which are the ones that matter.

1. The off-block projector equation was derived as $(\partial_a U)_{ji} + (u_j - u_i + N_j)X_{ji} = 0$, dropping the term $-X_{ji}N_i$. The correct equation is $\mathcal{S}_{ji}(X_{ji}) = -(\partial_a U)_{ji}$ with $\mathcal{S}_{ji}(X) = (u_j - u_i)X + N_jX - XN_i$. The closed form $D_a\mu_i = [C_{a,i}, S_i]$ therefore holds only when the block itself is semisimple, and the blocks carrying ν_6 are exactly the ones where it is not; in the draft's cubic block $J_0 \neq 0$.
2. The block was modelled as $z\partial_z - z^{-1}U_i + \mu_i$. After formal decoupling a nonsemisimple block carries higher powers of z , and its exponents depend on them: in the cubic block the z^2 coefficient's entry $-8/81$ enters $\rho^2 + \rho + 5/36 = 0$, and deleting it moves the exponents from $\pm 1/6$ to $\pm 1/18$. So the isomonodromy criterion stated there is a statement about semisimple blocks and is not a reduction of the residual problem. It is restated with that hypothesis.

The simple-block theorem survives, with one wording repair: the formal solution of a simple block is $e^{-u_i/z}$ times a single-valued formal unit, not times a constant, since decoupling can generate higher z -terms there too. The monodromy conclusion is unaffected.

From the second version to the third

Both changes come from running the three checks against the sources.

1. The framing route of Section [sec:hyzz] was called the probable repair. It is not a repair at all: its governing theorem lives at the large-radius limit point; its blowup case, though stated with existence in Theorem 5.22 and uniqueness in Theorem 5.24, is not derived from Theorem 4.34, whose hypotheses do not cover the unequal block sizes, and its existence is referred to the earlier decomposition work; and the gauge it supplies lands at the shifted base point the bulk gauge exists to undo.

2. The second version wrote that the unbounded negative loop order is the draft’s own choice rather than something the comparison theorems impose. That is wrong, and it was the ground for the recommendation. Iritani–Koto transport along the same object, their fundamental solution M of Section 5.8, with the same per-bulk-degree z^{-1} -polynomiality the draft derives in (4.1a).

Two findings were gained in exchange, and they narrow the problem rather than restate it: the comparison isomorphisms need no receiver, and the residual gap is one displacement statement. Both are in Section [sec:hyzz].

From the first version to the second

For readers of the first version, these are the substantive changes.

1. The Constants Lemma was stated without the constant-coefficient hypothesis and is false without it; see Remark [rem:constants-fails]. The framed-operator characterization inherits the hypothesis.
2. The scissors were only implicit. The transport theorem has no verified instance in the application, and that now appears where the theorem does.
3. The claim that the criterion is scale-invariant, and the inference that the separating parameter L cannot be tuned, were both wrong. Tuning L down is now the first proposed route.
4. The cubic-endpoint bullet compared two different normalizations. Corrected above.
5. The finite-tensor objection to the receiver was wrong twice over: the pro-Laurent gauge lies in the Hahn factor alone, and the splitting gauge’s coefficients lie in one finite extension of $H_{0,j}$. It is replaced by the gauge-relation argument.
6. The remark that a tensor product of two fields is not a domain was asserted as a general principle, which is false. The correct statement is that $\mathcal{R}_j$ need not be one.
7. The obstruction is now given in its order-free form, which answers two of the questions the first version left open: no completion helps, and the criterion is not an artifact of the ordered receiver.
8. Smaller repairs: the freeness proof (the relevant quotient group has torsion), the commutation of σ with the derivation, the inverse of M , the range of the exponents, the ramification factor in the splitting rate, and the divisor-tagging observation, which is new.